Sample Bridge Set 01: Binomial and Poisson

An example of what the `bridge-problems` skill produces, for a **fictional** intro-probability course (every problem is original). It builds toward a midterm's "count the successes" and "count the events" questions. Every numeric answer is checked by `verify_01_binomial_poisson.py`.

> **How to use this.** Each block: **read "Learn it" → problems in order → the past-test questions it unlocks.** Work on paper; open a hint only after ~5 minutes stuck; open the solution only once you have an answer.
> **Levels:** L1 warm-up · L2 two steps or a twist · L3 set it up yourself · L4 test-style.
> **Skip rule:** both L2s right without hints → jump to the L4.

What the (fictional) past tests ask

QUESTION TYPE	MIDTERM 2024	MIDTERM 2025	ASSIGNMENT 2	BLOCK
Binomial: "exactly k" / "at least one" / mean and variance	✓	✓	✓	A Must
Poisson: rate per interval, scale the rate	✓	✓		B Must
Binomial → Poisson approximation		✓		B4 Should
Geometric: "first success on trial k"	✓			not in this sample Skip-if-short

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Study plan (2 hours)

WHEN	DO	TIME
Tonight	Block A: Learn it → A1–A5 (skip rule after A3)	~1 h

Tomorrow	Block B: Learn it → B1–B4, then one past midterm under a timer and mark it	~1 h

Block A: Binomial Must

Unlocks: every "exactly / at least / expected number of successes" question.

Learn it

Big idea. You repeat the same yes/no trial *n* times, each trial succeeds with the same probability *p*, and the trials don't affect each other. The binomial distribution tells you how likely each total number of successes is, like hits in a hitter's at-bats.

Formulas. $P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n - k}$, where C(n, k) = n! / (k!(n − k)!) counts which trials succeed. $E[X] = np$, $Var(X) = np(1 - p)$.

Worked example. A 25% free-throw shooter takes 3 shots. P(exactly 1 make) = C(3, 1) · $0.25^1 \cdot 0.75^2$ = 3 × 0.25 × 0.5625 = **0.4219**.

Traps. (1) Forgetting C(n, k): there are several orders the successes can come in. (2) "At least one" by adding up many terms instead of 1 − P(none). (3) Using the SD where the variance is asked for (or the reverse).

Recipe.

1. Check: fixed n, two outcomes, same p, independent.
2. Write n, p and which k values the question wants.
3. "At least" → use the complement if it has fewer terms.
4. Plug in, then sanity-check (0 ≤ P ≤ 1; mean np is reasonable).

L1 A1. Two hits binomial pmf

A .300 hitter gets 4 at-bats. Treat at-bats as independent. What's the probability of exactly 2 hits?

Hint

n = 4, p = 0.3, k = 2. How many orders can 2 hits come in?

Solution

Binomial with n = 4, p = 0.3. $P(X = 2) = C(4, 2) \cdot 0.3^2 \cdot 0.7^2 = 6 \times 0.09 \times 0.49 =$ **0.2646**. Check: the mean is 1.2 hits, so 2 hits should be fairly likely but under 50%. It is.

L2 A2. At least one hit complement

Same hitter, 4 at-bats. What's the probability he gets at least one hit?

Hint

"At least one" has four cases. Its opposite has one.

Solution

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.7^4 = 1 - 0.2401 =$ **0.7599**. Check: adding P(1) + P(2) + P(3) + P(4) gives the same number, with four times the work.

L2 A3. A week of at-bats mean and SD

Over a 20-at-bat week, what's his expected number of hits, and the standard deviation?

Hint

Two formulas, then one square root.

Solution

$E[X] = np = 20 \times 0.3 =$ **6 hits**. $Var(X) = np(1 - p) = 20 \times 0.3 \times 0.7 = 4.2$, so $SD = \sqrt{4.2} \approx$ **2.05** hits. Check: the SD is in hits, like the mean. Variance is in hits^2, so it isn't the spread you'd quote.

L3 A4. Strikes in a sequence set it up yourself

A pitcher throws strikes on 62% of pitches. In a 5-pitch sequence, what's the chance at least 3 are strikes?

Hint

Here the complement has 3 terms too, so just add k = 3, 4, 5.

Solution

n = 5, p = 0.62. $P(3) = 10 \times 0.62^3 \times 0.38^2 = 0.3441$; $P(4) = 5 \times 0.62^4 \times 0.38 = 0.2808$; $P(5) = 0.62^5 = 0.0916$. Sum = **0.7165**. Check: the mean is 3.1 strikes, so "at least 3" should be more likely than not. It is.

L4 A5. First-pitch strikes (test-style) all of block A

A reliever throws a first-pitch strike to each batter with probability 0.6, independently. He faces 8 batters.

a. P(exactly 5 first-pitch strikes)?

b. P(at most 2)?

c. Mean and variance of the count.

d. What's the smallest number of batters he must face so the chance of *at least one* first-pitch ball is at least 0.99?

Hint

(d): "at least one ball" = 1 − P(all strikes) = $1 - 0.6^n$. Solve for n with logs, then round the right way.

Solution

(a) $C(8, 5) \cdot 0.6^5 \cdot 0.4^3 = 56 \times 0.07776 \times 0.064 =$ **0.2787**.

(b) P(0) + P(1) + P(2) = 0.00066 + 0.00786 + 0.04129 = **0.0498**.

(c) Mean = 8 × 0.6 = **4.8**; variance = 8 × 0.6 × 0.4 = **1.92**.

(d) Need $1 - 0.6^n \geq 0.99$, so $0.6^n \leq 0.01$, so $n \geq \ln 0.01 / \ln 0.6 = 9.02$. Round *up*: **n = 10**. Check: $0.6^9 = 0.0101$ (just misses) and $0.6^{10} = 0.0060$ (works). Dividing by ln 0.6, a negative number, flips the inequality.

Block B: Poisson Must

Unlocks: "events happen at an average rate" questions, and approximating a binomial with large n and small p.

Learn it

Big idea. When events happen at random at a steady average rate (home runs per game, errors per inning), the count in a fixed window is Poisson. There's no fixed n: just the average count λ for *that* window.

Formulas. $P(X = k) = e^{-\lambda}\lambda^k / k!$; $E[X] = Var(X) = \lambda$. Rate scales with the window: λ for 3 games = 3 × λ per game.

Worked example. A team averages 2 stolen bases a game. P(exactly 2 in a game) = $e^{-2} \cdot 2^2 / 2!$ = 0.1353 × 2 = **0.2707**.

Traps. (1) Not rescaling λ to the window the question asks about. (2) 0! = 1, so P(0) = $e^{-\lambda}$. (3) Using a Poisson where n is small and p isn't (then it's binomial).

Recipe.

1. Find the rate and the window; λ = rate × window length.
2. "At least" → 1 − (the few terms below).
3. Plug in and check that the mean (= λ) is reasonable.

L1 **B1. A quiet game** Poisson pmf at 0

A team averages 1.2 home runs a game. What's the probability they hit none in a game?

Hint

k = 0 makes the formula very short.

Solution

P(0) = $e^{-1.2}$ = **0.3012**. Check: about a third of games with no homer feels right for a 1.2 average.

L2 B2. A three-game series rescale + complement

Same team, a 3-game series. What's the probability of at least 2 home runs in the series?

Hint

New window, new λ. Then 1 − P(0) − P(1).

Solution

$\lambda = 3 \times 1.2 = 3.6$. $P(0) + P(1) = e^{-3.6}(1 + 3.6) = 0.02732 \times 4.6 = 0.1257$. So $P(X \geq 2) =$ **0.8743**. Check: with 3.6 expected, fewer than 2 should be rare. It is (about 13%).

L3 B3. A clean stretch set it up yourself

A defence makes 0.5 errors a game on average. The coach brags about 4 straight games without an error. How likely was that by chance alone?

Hint

One window of 4 games, or four 1-game windows multiplied: both work.

Solution

$\lambda = 4 \times 0.5 = 2$, so $P(0) = e^{-2} =$ **0.1353**. Equivalently $(e^{-0.5})^4$, because the games are independent. Check: the two methods agree, which is a nice property of the Poisson.

L4 **B4. Hit batters (test-style)** binomial vs. Poisson approximation

A pitcher hits the batter on 1% of plate appearances and faces 250 over a season.

a. Exact P(at most 1 hit batter), using the binomial.

b. The same with the Poisson approximation.

c. How far apart are they?

d. P(5 or more), exactly.

Hint

The Poisson approximation uses $\lambda = np$. Compare the two answers to 4 decimals.

Solution

(a) $0.99^{250} + 250 \times 0.01 \times 0.99^{249} = 0.08106 + 0.20469 =$ **0.2858**.

(b) $\lambda = 250 \times 0.01 = 2.5$: $e^{-2.5}(1 + 2.5) =$ **0.2873**.

(c) A difference of **0.0015**: large n and small p make the Poisson a very good shortcut.

(d) $1 - P(X \le 4) =$ **0.1078** (the Poisson gives 0.1088). **Check:** both models have mean 2.5, and the binomial variance (2.475) is just under the Poisson's (2.5).